

Mechanical work analysis for the CL-PFU database

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```
# Set the version we want to use for this report.  
vrsn_str <- "v3.0b1"
```

1 Introduction

This report investigates world aggregate Mechanical work. The questions were raised by Tiago Domingos regarding Figure 7 of Brockway et al. (2024).

2 Data

The first step is to gather relevant data from the database and save locally for convenience.

```
conn <- PFUPipelineTools::get_scratchmdb_conn()  
on.exit(DBI::dbDisconnect(conn))  
  
world_mats <- PFUPipelineTools::pl_filter_collect(  
  version_string = vrsn_str,  
  db_table_name = "PSUTReAllChopAllDsAllGrAll",  
  Country == "World",  
  conn = conn,  
  collect = TRUE)  
world_mats |>  
  saveRDS(file = "world_mats.rds")  
  
DBI::dbDisconnect(conn)
```

3 Analysis

3.1 Recreate Figure 7 from the database paper

Note that all analyses are performed at the world level, with gross useful energy and exergy. (Gross includes useful stage energy and exergy consumed by final demand *and* energy industry own use. These data are taken from the $\mathbf{Y}$ and $\mathbf{U}_{EIOU}$ matrices.) This analysis employs v3.0b1 of the CL-PFU database, while the paper (and Figure 7) used v1.2. Some small differences can be expected.

The first step in the analysis is to create Figure 1, a graph similar to Figure 7 from the database paper. By this step, we double-check the data we obtained from the database. Here, we're using aggregated (Grouped) data. We see that Mechanical work (MechW) appears nearly constant, as observed (and questioned) by Tiago.

Next, we create a graph using Specified data (the most raw data) to confirm equivalence of Specified data and the Grouped data of Figure 1 (and Figure 7 of the database paper). The Specified data are required to

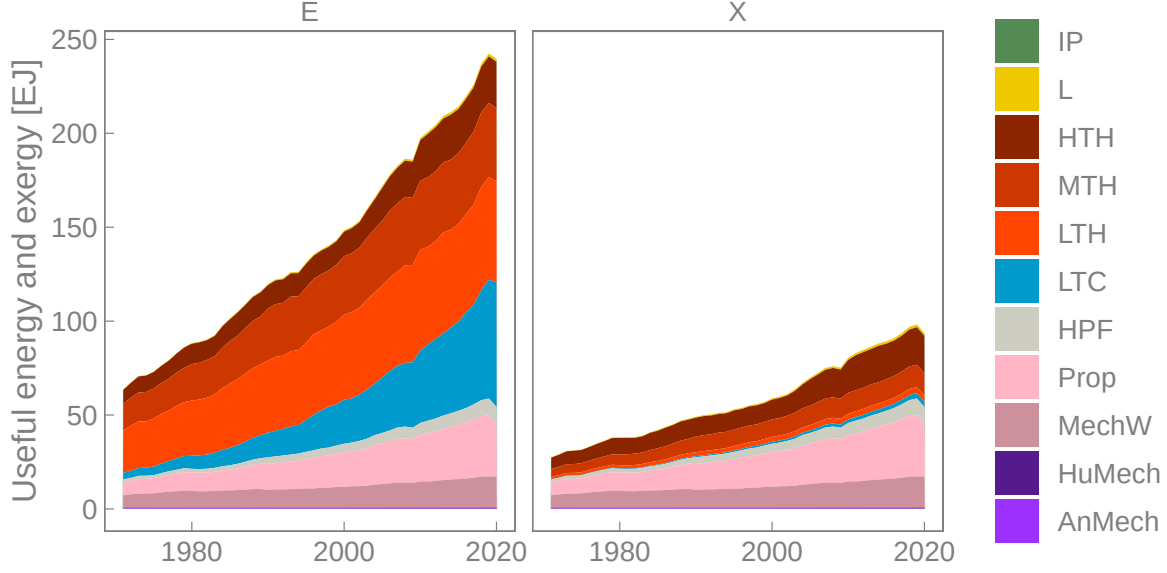


Figure 1: Remake of Figure 7 of the database paper using pre-aggregated (i.e., Grouped) data.

disaggregate Mechanical work. Figure 2 is the same as Figure 1, indicating (a) that we have consistency in our data and (b) that we can use the Specified data to disaggregate the Mechanical work time series.

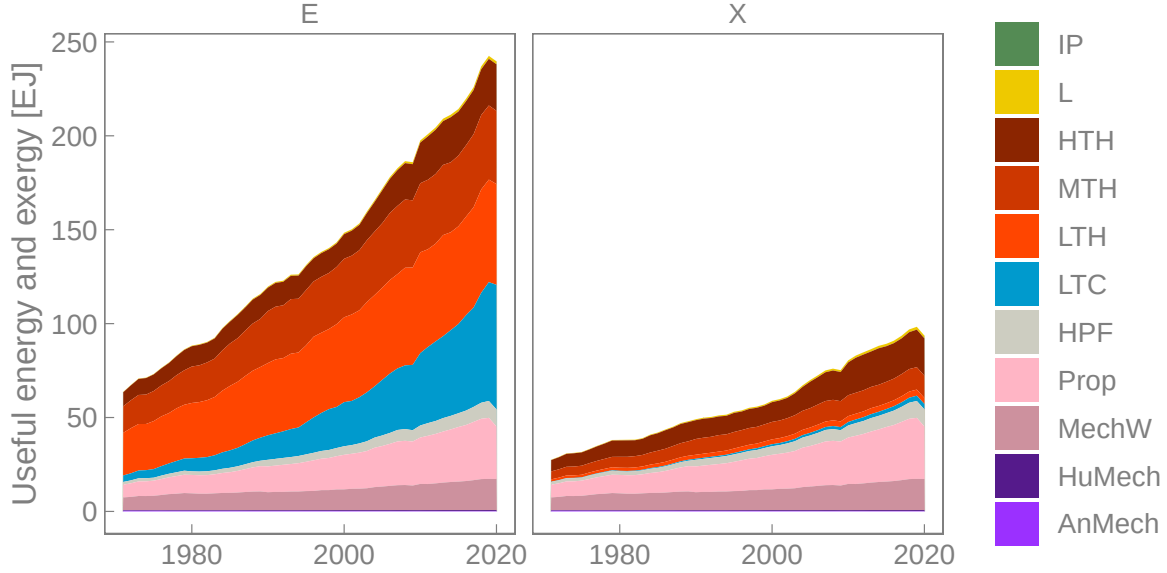


Figure 2: Remake of Figure 7 from the database paper starting from Specified data.

Next, we focus on Mechanical work.

3.2 Mechanical work

The Mechanical work values for 1971 and 2020 are shown in Table 1. Mechanical work is not *actually* constant over time. Over the 50-year period, Mechanical work grows 145.2% for a compound average annual growth rate of 1.8 %/yr.

The Mechanical work time series are shown in Figure 3.

Table 1: Mechanical work values (in EJ) for 1971 and 2000.

Year	E	X
1971	6.76	6.76
2020	16.58	16.58



Figure 3: World mechanical work.

Next, we disaggregate Mechanical work into subcategories. In the CL-PFU database, we calculate aggregate Mechanical work by summing KE, MD, MF, and MacW. Figure 4 shows that Mechanical work (MechW) is growing (in total), but Mechanical drive (MD) is declining slightly while Machine work (MacW) and Kinetic energy (KE) grow through time. Material fracture (MF) is insignificant over the timespan of the CL-PFU database.

Now, we plot disaggregated Mechanical work at the same scale as Figure 7 from the database paper. Here, we see that Mechanical work appears quite flat, as Tiago noticed, despite its growth.

Table 2 shows numerical values for disaggregated Mechanical work in 1971 and 2020.

Table 2: Mechanical work components (in EJ) for 1971 and 2000.

Year	KE	MacW	MD	MF
1971	0.82	1.41	4.53	0.01
2020	5.42	3.96	7.17	0.03

3.3 Mechanical drive (MD)

The largest subcategory of Mechanical work (MechW) is Mechanical drive (MD), and MD is quite flat over time. To gain further insight, we extract, from the $\mathbf{V}$ (make) matrix when the last stage is Useful, machines that produce Mechanical drive (MD). The machines that produce MD are Domestic appliances, Natural gas turbines, and Static diesel engines. Figure 6 shows the evolution of world-total outputs from those three machines.

Figure 6 shows that Static diesel engines are producing less MD over time, a decline that is partially offset by growing MD production by Domestic appliances. Static gasoline engines provide a very minor contribution to

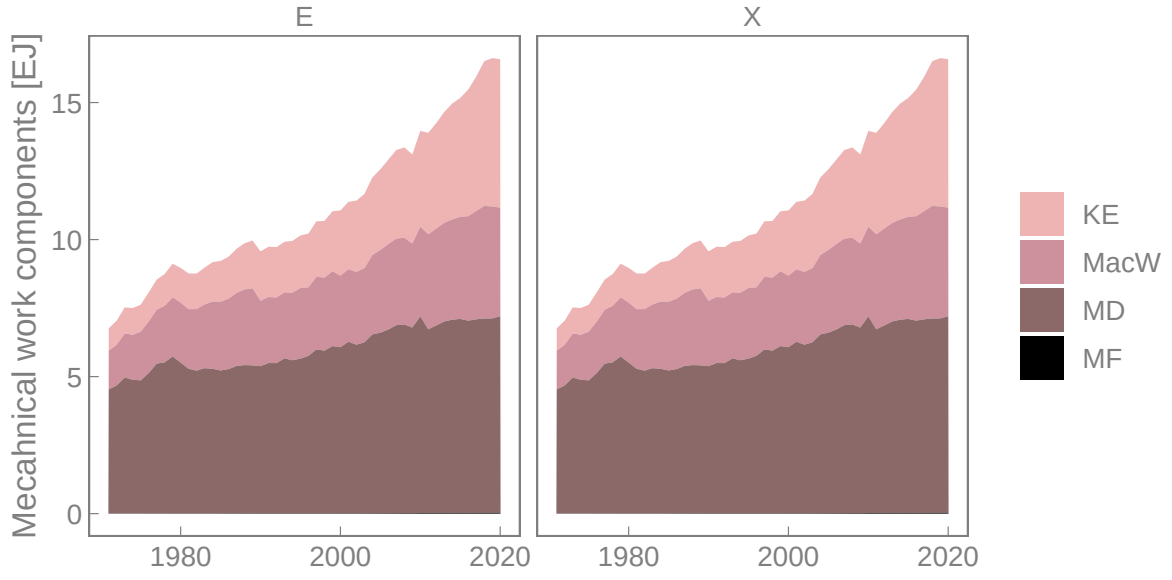


Figure 4: Disaggregation of mechanical work.

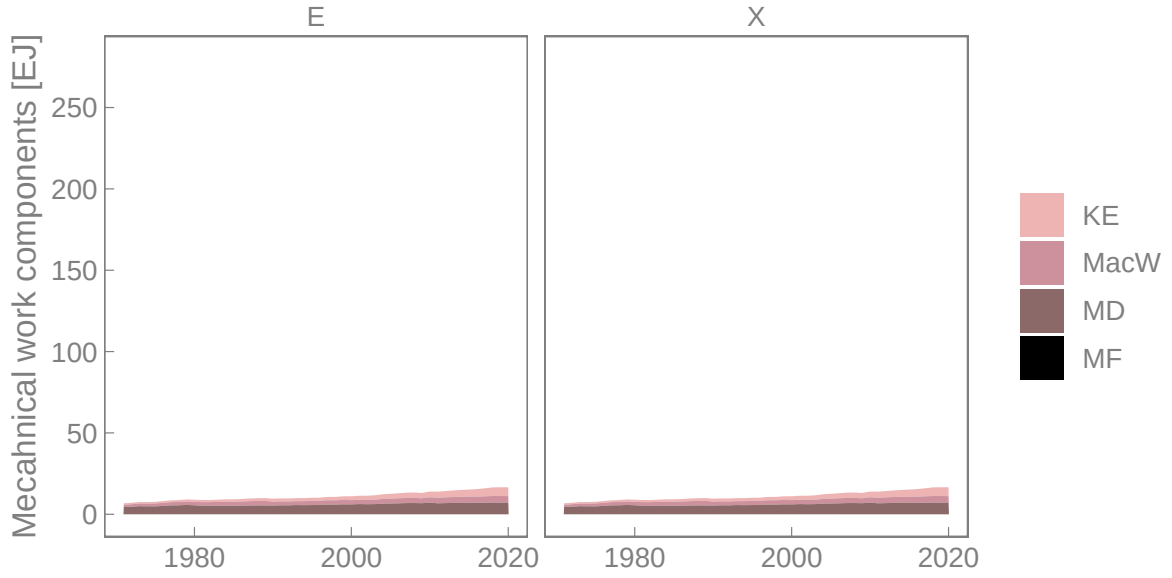


Figure 5: Mechanical work at same scale as Figure 7 from the database paper.

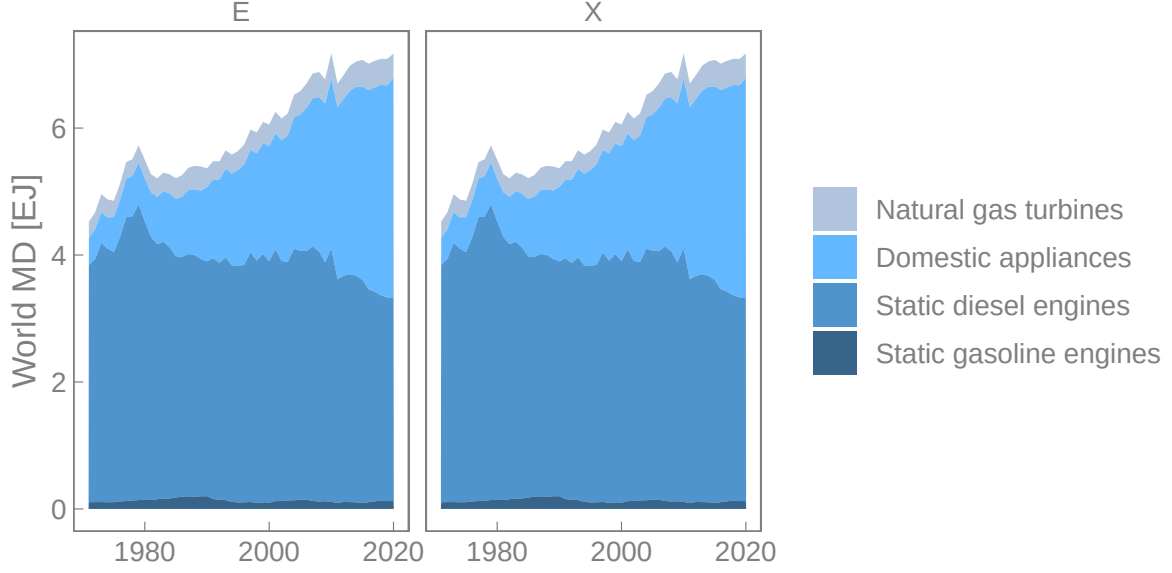


Figure 6: World mechanical drive (MD) by source.

total Mechanical drive.

3.4 Inputs to MD machines

Domestic appliances, Natural gas turbines, and Static diesel engines consume energy to make MD. We extract, from the $\mathbf{U}$ (use) matrix when the last stage is Useful, energy and exergy inputs to machines that make Mechanical drive (MD). Figure 7 shows the evolution of world-total inputs to those machines.

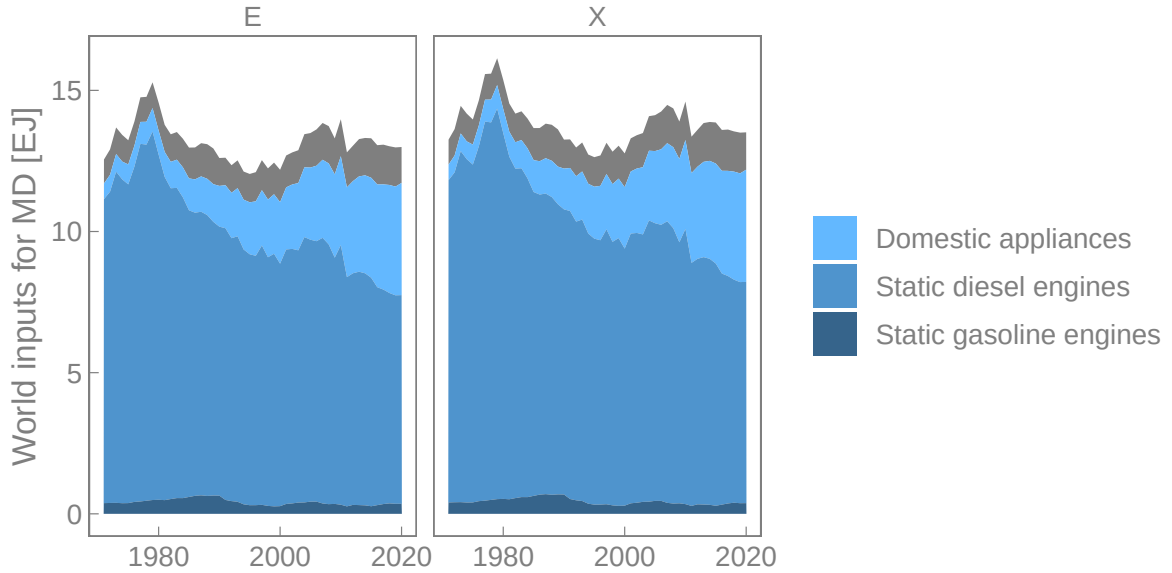


Figure 7: World inputs for mechanical drive (MD) by machine.

Figure 7 shows that input energy for Static diesel engines is declining over time. With declining MD output from Static diesel engines, these results suggest declining usage of Static diesel engines over time.

Both input energy to and output energy from Domestic appliances is growing over time, suggesting growing use of Domestic appliances over time.

3.5 Efficiencies of MD machines

It is also instructive to look at the efficiency of the machines that produce MD (Domestic appliances, Natural gas turbines, and Static diesel engines). Efficiency can be calculated by dividing MD machine outputs (found in the **V** matrix) by MD machine inputs (found in the **U** matrix). Figure 8 shows the evolution of those efficiencies.

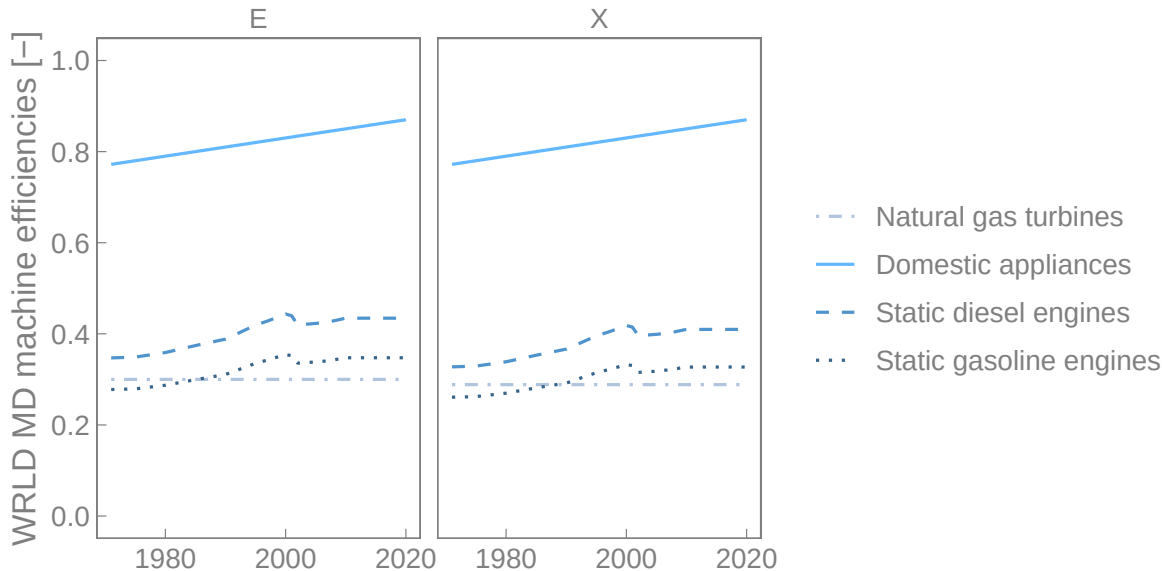


Figure 8: World-level efficiencies of machines that produce MD.

A few observations can be made.

- There is an interesting inflection in World engine efficiencies around 2000 that should be investigated.
- The efficiency of Static diesel engines rises from 1971–2020, thus MD output declines more slowly than energy inputs.
- The energy and exergy output from (i.e., produced by) MD machines is identical. MD machine efficiencies are identical across energy and exergy for Domestic appliances, because the energy and exergy of input electricity is identical. Most Domestic appliances are powered by Electricity.
- The efficiency of Static diesel and gasoline engines is slightly different between energy and exergy due to the slight difference in energy and exergy of input fossil fuels. Because the exergy is larger than energy for fossil fuels, exergy efficiency is slightly lower for Static diesel and gasoline engines.

3.6 Consumption of MD by sectors

We can interrogate the **Y** (final demand) matrix for usage of MD by various sectors. Figure 9 shows the evolution of MD consumption.

The Iron and steel sector is largely decreasing in MD consumption, while Residential exhibits large increases in MD consumption.

Table 3 shows differences in MD consumption over the 60-year period from 1971 to 2020.

4 Conclusion

It is true that Mechanical work (MechW) grows more slowly than other categories of useful energy and exergy. However, Mechanical work *is growing* at 1.8 %/yr, 0.56 of the compound average annual world GDP growth rate (3.3 %/yr) over the same period.

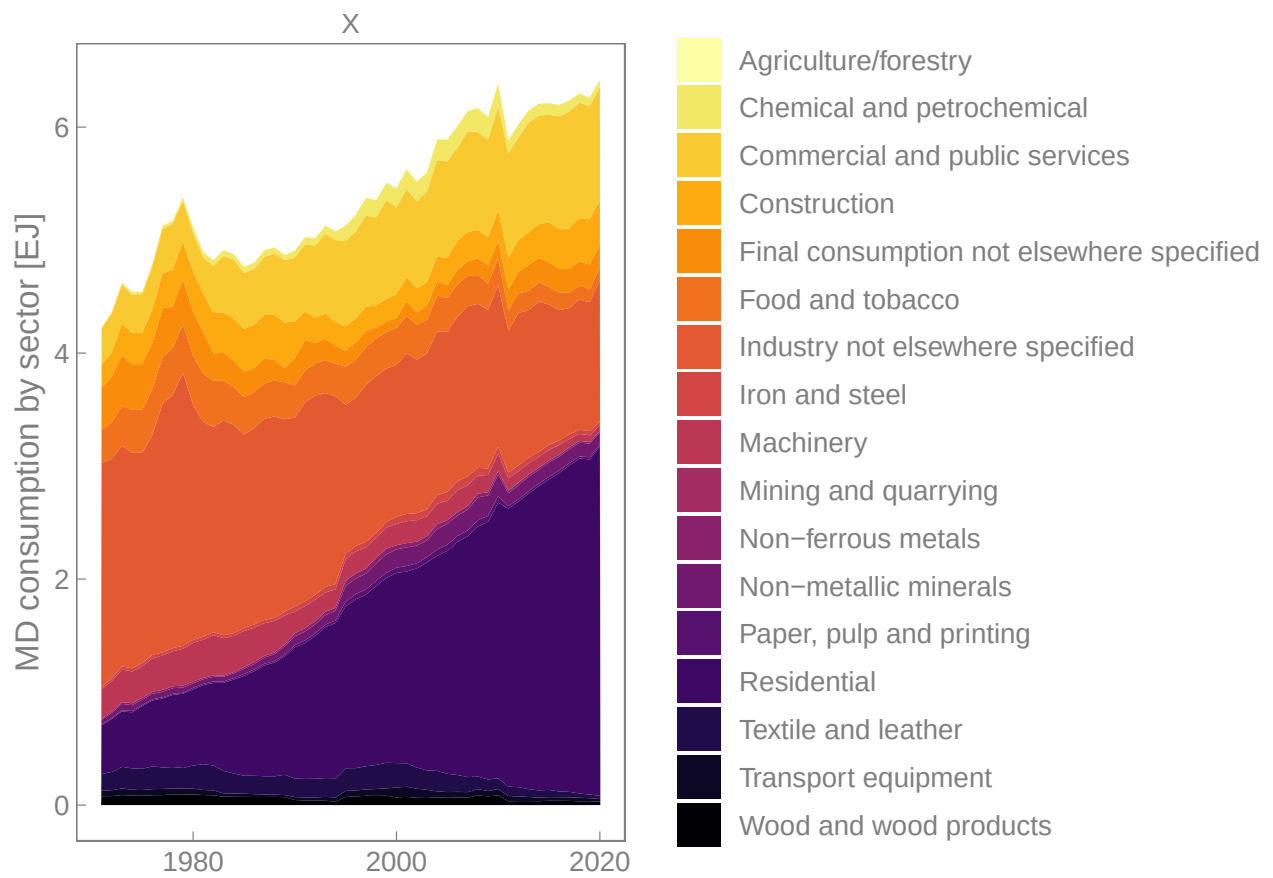


Figure 9: World-level MD consumption by sector.

Table 3: Mechanical drive (MD) exergy values (in EJ) for 1971 and 2000, arranged by difference.

Sector	1971	2020	Difference
Residential	0.43	3.09	2.66
Commercial and public services	0.32	1.01	0.69
Construction	0.20	0.39	0.19
Non-metallic minerals	0.04	0.10	0.07
Chemical and petrochemical	0.01	0.07	0.06
Iron and steel	0.02	0.03	0.02
Paper, pulp and printing	0.01	0.02	0.01
Non-ferrous metals	0.01	0.01	0.00
Transport equipment	0.05	0.02	-0.02
Wood and wood products	0.08	0.03	-0.05
Textile and leather	0.15	0.03	-0.12
Final consumption not elsewhere specified	0.37	0.21	-0.17
Food and tobacco	0.29	0.11	-0.19
Machinery	0.26	0.05	-0.21
Industry not elsewhere specified	1.98	1.25	-0.73
Agriculture/forestry		0.00	
Mining and quarrying		0.00	

The largest category of MechW is Mechanical drive (MD) produced by Domestic appliances, Natural gas turbines, and Static diesel engines. The largest producer of MD is Static diesel engines, but Static diesel engines produce less energy in 2020 than in 1971, despite rising efficiency, causing decreasing energy inputs due to less usage over time. Industrial usage of MD are declining over time, offsetting increases in consumption of MD by the Residential sector.

References

Brockway, Paul E, Matthew Kuperus Heun, Zeke Marshall, et al. 2024. "A Country-Level Primary-Final-Useful (CL-PFU) Energy and Exergy Database: Overview of Its Construction and 1971–2020 World-Level Efficiency Results." *Environmental Research: Energy* 1 (025005): 1–29. <https://doi.org/10.1088/2753-3751/ad4e39>.